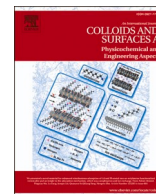




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Molecular simulations of multivariate competitive adsorption of CH₄, CO₂ and H₂O in gas-fat coal

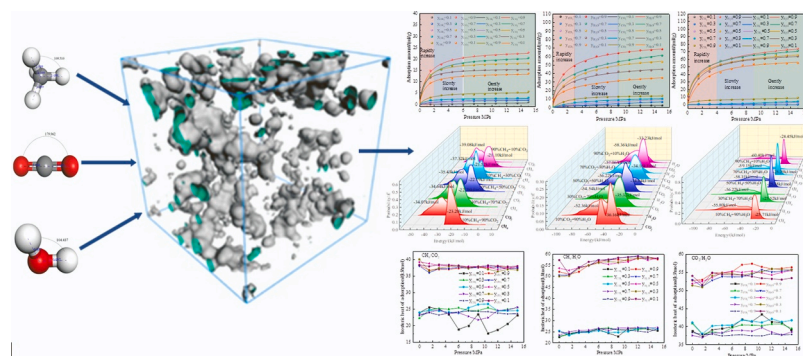
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GRAPHICAL ABSTRACT



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ABSTRACTS

There is a physical process of CH₄, CO₂ and H₂O multi-gas competitive adsorption in the process of injecting flue gas or CO₂ to enhance coalbed methane recovery. In order to clarify the microscopic mechanism of the competitive adsorption of CH₄, CO₂ and H₂O, a macromolecular structure model of Jixi gas-fertilized coal was constructed in this paper, and the competitive adsorption behaviors of binary and ternary mixed-component gases with different molar volume fractions were investigated by using the Giant Canonical Monte Carlo (GCMC) and Molecular Dynamics (MD) methods at the temperature of 293.15 K. To analyze the laws of pressure, volume fraction and competition between different gas components for the effect of adsorption amount, isosteric heat of adsorption, interaction energy, energy distribution, probability density distribution, and adsorption selectivity coefficient. The results showed that under the same conditions, the adsorption amounts from high to low were H₂O>CO₂>CH₄. The adsorption amounts of each gas component increased with the increase of volume fraction. The adsorption selectivity coefficient decreases with increasing equilibrium pressure and increasing molar volume fraction of strongly adsorbed gas. CO₂ and H₂O occupy the adsorption sites of CH₄, making the CH₄ adsorption space smaller and the adsorption sites more concentrated. The interaction energy increases with the increase of the molar fraction of strongly adsorbed gases in the multifunctional system.

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1. Introduction

There is a physical process of competitive adsorption of CH₄, CO₂, and H₂O in the process of improving coalbed methane recovery by injecting flue gas or CO₂. In order to clarify the micro mechanism of competitive adsorption of CH₄, CO₂, and H₂O on the gas-rich coal macromolecular model, the feasibility of improving coalbed methane recovery by injecting flue gas or CO₂ while achieving CO₂ geological storage is explored.

A large number of experts and scholars have conducted a lot of experimental and simulation studies on the adsorption and diffusion mechanism of CBM in coal. Du [1] carried out a simulation study on the adsorption of a mixture of CH₄, CO₂ and N₂ on montmorillonite using MS software, confirming that CO₂ can stimulate CH₄ desorption by molecular exchange, that lowering the temperature increases the selectivity, and that a low-temperature gas injection is more conducive to triggering the desorption behavior of the adsorbed gases. Iddphonce [2] confirmed the scientific and feasibility of CO₂-ECBM technology through experiments and simulations. Hu Yu [3] explored the effect of coal oxidation on the adsorption characteristics of CH₄ and CO₂ and the adsorption conformation of molecules at different functional groups in coal, and the results showed that oxidation can change the pore structure of coal, resulting in an increase in both gas adsorption and diffusion coefficient, as well as an increase in the adsorption selectivity of the gas. The distance between gas molecules and functional groups determines the expression of interaction forces. There exists a critical distance between gas molecules and functional groups, which is smaller than the critical distance as repulsive force and larger than the critical distance as attractive force. Li Shugang [4] investigated the adsorption behavior of CH₄, CO₂ and N₂ in coal from a thermodynamic point of view, and showed that the adsorption energy has a positive correlation with the heat of adsorption, and that the adsorption potential is affected by the enthalpy of adsorption, the amount of adsorption and the entropy of adsorption. Bai [5] in order to study the dynamical mechanism of CH₄ replacement by CO₂ and N₂, used a Giant Canonical Monte Carlo simulation method to investigate the desorption behavior of CH₄ after injection of CO₂ and N₂ at different temperatures, and the results showed that CO₂ and N₂ mainly displaced methane by occupying the adsorption sites of CH₄. The average relative concentrations and average velocities of the three gases in the vacuum layer were distributed CH₄ > CO₂ > N₂. They increased gradually with the increase of temperature, and the diffusion activation energy of CH₄ decreased by 20.53% in the model with CO₂ injection, and the displacement effect of injecting CO₂ was better than that of N₂. Kumar [6] and others carried out a series of gas injection replacement experiments to comparatively analyze the present of gas injection replacement coalbed methane technology from three aspects: gas injection location, gas injection method and gas injection type. Zhou Junping [7] studied the physical processes such as adsorption diffusion percolation of CH₄, CO₂, N₂ and other multiple gases in coal during the process of carbon dioxide sequestration, and used molecular simulation to study the effects of temperature, pressure and pore structure on the adsorption of multiple gases, which provided theoretical references for the optimization of the on-site process.

Busch [8] conducted experiments on the adsorption diffusion of a binary mixture of CH₄/CO₂ in coal, which showed that the adsorption rate of CO₂ was consistently 2–3 times higher than that of CH₄. Rana-thunga [9] conducted CO₂ driven coal bed methane experiments to study the applicability of CO₂ enhanced coal bed methane recovery, which showed that injection of CO₂ has a higher coal bed methane recovery rate compared to natural pumping, and supercritical CO₂ has superior performance in the process of improving CH₄ recovery rate, and the permeability enhancement performance is independent of coal rank. Zeng [10] used an adsorption-strain coupling model to simulate the coalbed methane mining process, and the results showed that it takes 1800 days to replace 90% of the methane considering the deformation of the coal and the change of the permeability, but it is difficult for the

macroscopic experiments to explore the microscopic role of the replacement process. Liu [11] used defective and non-defective graphene to represent organic matter in coal and shale, and investigated both with CO₂ adsorption by density functional theory, which showed that defective graphite is more likely to adsorb CO₂. Zhao [12] investigated the behavior of binary adsorption and diffusion of CO₂/CH₄ using molecular simulation to provide an optimal injection depth for CO₂-ECBM. Dang [13] investigated the adsorption behavior of CO₂ and CH₄ in low-rank coal by using DFT and molecular simulation, and oxygen-containing functional groups and nitrogen-containing functional groups were more adsorbent to CO₂ and CH₄, and high temperatures inhibited the adsorption behavior.

Zhang Songhang [14] The pore structures of three anthracite coals from Qingshui Coalfield were explored, and competitive adsorption and desorption experiments of CH₄ and CO₂ gas mixtures were carried out. The study showed that the pore structure has an important influence on the gas adsorption and transportation in coal, the ultra-micro-pores (<2 nm) are the significant adsorption sites, and the desorption hysteresis index is smaller when the pore size > 100 nm has more pore development. The desorption hysteresis phenomenon is mainly caused by the opposite migration direction of the adsorption and desorption processes and the misalignment of the coal matrix contraction. The (CO₂/CH₄) separation index decreased with the increase of adsorption equilibrium pressure. The equivalent macroscopic diffusion coefficient of the gas mixture increased with increasing CO₂ content. Zhang hongyang [15] The replacement of CH₄ by a CO₂-N₂ gas mixture with different components was simulated using Giant Canonical Monte Carlo, and the recovery of CH₄, the storage of CO₂, and the pore deformation characteristics were investigated. The injection of CO₂-N₂ can reduce the formation swelling, the recovery of CH₄, and the adsorption deformation of CO₂ with the increase of pore diameter. The addition of N₂ can effectively reduce the pore swelling. Wang Xiaoqi [16] Giant Canonical Monte Carlo to study the competitive adsorption behavior of CO₂ and CH₄. CO₂ showed high competitive adsorption capacity, and the selectivity coefficient of CO₂/CH₄ decreased with the increase of temperature and pressure. Long Hang [17] use the microscopic adsorption mechanisms of mono-component, dicomponent, and tri-component of CH₄, CO₂, and N₂ in Wiser bituminous coal were investigated by GCMC and MD methods. The results showed that CO₂ had the highest adsorption capacity and N₂ had the lowest adsorption capacity, and the adsorption amount increased and the adsorption separation coefficient decreased with the concentration of CO₂ in the binary component. High concentration of CO₂ had a negative effect on competitive adsorption. CO₂ had the strongest interactions in the coal molecule, and the isosteric heat of adsorption was inversely proportional to the molar volume fraction of CO₂. Wu [18] in order to reveal the microscopic mechanism of the competitive adsorption of CO₂, O₂, and N₂, and to provide theoretical support for injecting coal combustion flue gas into the air-mining area to effectively store CO₂ and prevent fire. A macromolecular coal model was established, and the competitive adsorption of monomer, binary and ternary was investigated based on the GCMC and MD methods, and the results showed that CO₂ has greater electrostatic and van der Waals energies with coal, and the interaction energy with coal is greater than that of O₂ and N₂, so the adsorption of the three gases is: CO₂ > O₂ > N₂. Yves Gensterblum [19] conducted adsorption experiments on CH₄, CO₂ and H₂O from sub-bituminous to anthracite coals at different water contents from 303 to 350 K. It was shown that the adsorption capacity of dry coal was parabolic with increasing reflectance of the mirror group and linear for wet coal. Oxygen-containing functional groups are the main competing adsorption sites for CH₄, CO₂ and H₂O, and when water is present in the adsorbed phase, the adsorbed molecules are more localized, and the thermodynamic properties of the adsorbed molecules are dominated by adsorbate-adsorbate at high surface coverage. Alexej Merkel [20] experimentally investigated the competitive adsorption of two-component CH₄/CO₂ by coals of three different coal classes (sub-bituminous, high-volatile bituminous, and anthracite) at dry , relative

humidity of 53% and 97%. It was shown that the adsorption capacity of the coals for the gas mixture increased with decreasing moisture, most pronounced in the sub-bituminous coals. The selectivity of CO₂ adsorption for CH₄ decreased with increasing maturity and moisture, with selectivity coefficients ranging between 6 and 9. Li Jiawei [21] used a combination of molecular dynamics and Giant Canonical Monte Carlo to investigate the effects of moisture, salinity and ethane on the competitive adsorption properties of CH₄/CO₂. Moisture, salinity and ethane hindered CH₄ adsorption. Moisture, salinity and ethane promoted the increase of CO₂/CH₄ selectivity, which contributed to the improvement of the replacement efficiency of CO₂ for CH₄.

Experts and scholars have conducted a large number of experimental and simulation studies on CH₄ and CO₂ adsorption, revealing the competitive adsorption mechanism between coal and CH₄ and CO₂ single and binary components, and scholars have conducted a large number of simulation and experimental studies on coal adsorption of water molecules, revealing the adsorption characteristics of water in the coal seam, but there are fewer studies on the impact of water molecules on the adsorption characteristics of CH₄ and CO₂, lack of research from the molecular level to investigate the impact of different temperatures and pressures on the adsorption and diffusion characteristics of coal adsorption of CH₄, CO₂ and H₂O molecules, and the lack of studies on the microscopic mechanisms of adsorption for the competitive adsorption of CH₄, CO₂ and H₂O multifunctional mixed components. Therefore, the aim of this paper is to investigate the competitive adsorption behavior of binary (CH₄/CO₂, CO₂/H₂O, CH₄/H₂O) and ternary (CH₄/CO₂/H₂O) mixed-component gases with different molar volume fractions at a temperature of 293.15 K by means of Giant Canonical Monte Carlo (GCMC) and Molecular Dynamics (MD) methods. To analyze the laws of pressure, volume fraction and competition between different gas components for the effect of adsorption amount, isosteric heat of adsorption, interaction energy, potential energy distribution, probability density distribution, and adsorption selectivity coefficient.

2. Methodology

2.1. Molecule model

Coal is a kind of non-homogeneous porous material with complex physical and chemical structure, domestic and foreign experts and scholars use a variety of characterization and testing instruments to analyze the coal macromolecular structure, to obtain all kinds of functional groups and microcrystalline structure information of coal, and construct a large number of models of coal macromolecular structure. Fuch[22] first constructed a coal model in 1942, Given[23] in 1960, 1970 Wiser[24], 1984 American scholar Shinn[25] have constructed the coal macromolecular structure model, in this paper, using FTIR, XPS, ¹³C NMR, XPS, industrial, elemental analysis and other test methods, independently constructed the gas-fat coal macromolecule model, the molecular formula (C₂₁₂H₁₈₉O₃₈N₃S), the planar two-dimensional structure shown in Fig. 1.

The two-dimensional planar model of gas-fat coal was imported into MS software, saturated with hydrogen, and geometrically optimized, annealed, and dynamically optimized to finally obtain the macromolecular lowest energy configuration of gas-fat coal. The Amorphous Cell module was used to construct the three-dimensional cell structure of gas-fat coal, and 15 macromolecule minimum energy models of gas-fat coal were put into the cell to add three-dimensional periodic boundaries, and the parameters were set as shown in Table 1, and the molecular dynamics method was used to perform the geometric optimization, annealing optimization, and dynamic optimization on the cell structure of Jixi gas-fat coal to get the minimum energy configuration of the three-dimensional cell structure of Jixi gas-fat coal, as shown in Fig. 2.

The density of gas-fat coal macromolecule cell model is stable at 1.14 g/cm³, and the measured density of Jixi gas-fat coal is 1.17 g/cm³,

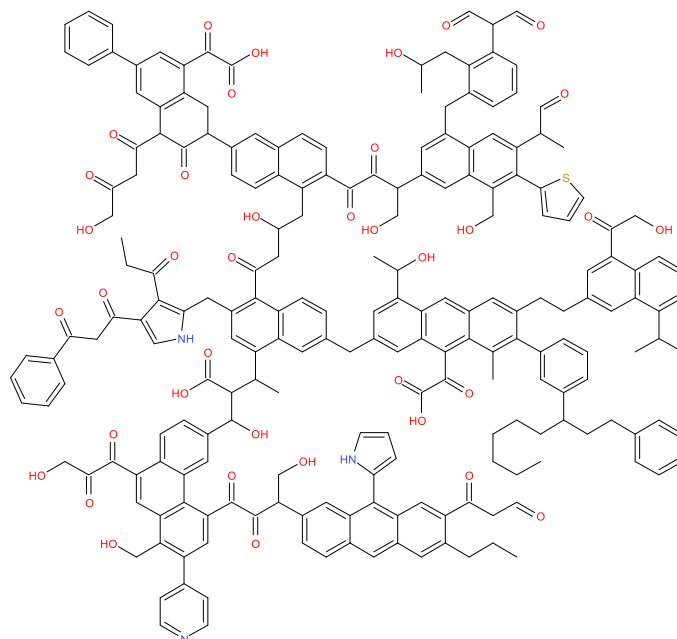


Fig. 1. Planar model of macromolecular structure of gas-fat coal.

and the simulated density is slightly lower than the measured value. On the one hand, due to the presence of quartz sand, kaolinite and other non-mineral substances in the coal, on the other hand, there are small molecules of organic matter in the coal macromolecule model, the modeling process does not take into account the non-mineral substances and small molecules of organic matter resulting in the simulated density is lower than the experimental density, it is considered that the gas-fat coal cell structure constructed in this paper is reasonable. The cell structure of Jixi gas-fat coal has the dimensions A=B=C= 4.09543 nm, $\alpha = \beta = \gamma = 90^\circ$, and the molecular formula C₃₁₈₀H₂₈₃₅N₄₅O₅₇₀S₁₅. The microporous distribution characteristics of Jixi gas-fat coal were investigated by using the Atom Volumes & Surfaces module, with the Connolly probe radius of 0.13 nm and the van der Waals' scanning factor of 1.0 Å. The microporous distribution of the model is shown in the following table. Å, the microporous distribution of the model is shown in Fig. 2.

Using Material Studio software to construct CH₄, CO₂ and H₂O molecules, CH₄ is tetrahedron structure, CO₂ is rectilinear triatomic structure, and H₂O is folded triatomic structure, the energy optimization of gas small molecule model to get the lowest energy model of CH₄, CO₂ and H₂O molecules, as shown in Fig. 3, the three gas molecules were optimized to get the final energy close to zero, the three adsorbate molecules have a large difference in terms of their physical properties, as shown in Table 2, which is an important reason of the different nature of gas-fat coal adsorbed on them.

2.2. Simulation methodology and parameter setting

The adsorption behavior of CH₄, CO₂ and H₂O molecules on the macromolecular structural model of Jixi gas-fat coal were investigated based on the Giant Canonical Monte Carlo and molecular dynamics

Table 1
Density simulation parameters.

Setting	Parameter	Setting	Parameter
Take	Construction	Force field	COMPASS
Quality	Medium	Charges	Forcefield assigned
Density	1.14 g/cm ³	Electrostatic	Atom based
Temperature	298 K	van der Waals	Atom based

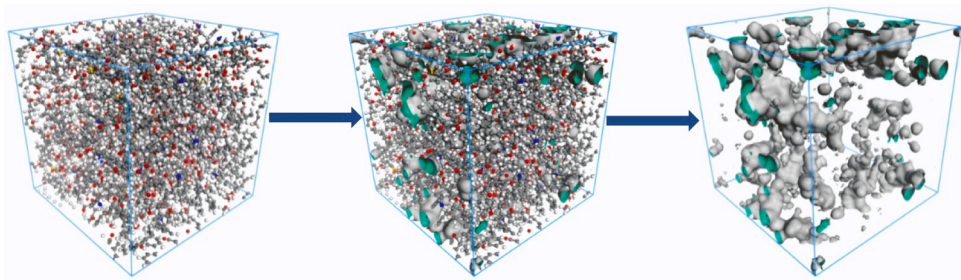


Fig. 2. 3D macromolecular pore characteristics of gas-fat coal.

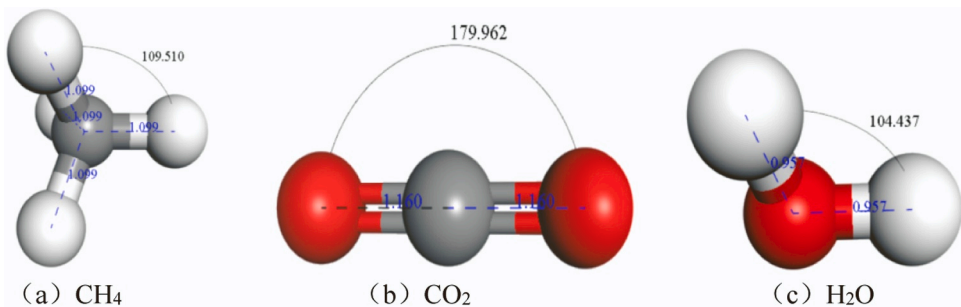


Fig. 3. CH₄, CO₂ and H₂O small molecule models.

methods. The Sorption module was used, and the parameter settings of the MS software are shown in Table 3.

The adsorption volume (moleculars/u.c) obtained from the software simulation was converted to the experimental and engineering adsorption volume (ml/g), which can be expressed as:

$$Q = \frac{N}{M} \tag{1}$$

In formula: Q is the gas adsorption amount, mol/g; N is the number of adsorbed gases, moleculars/u.c; M is the molar mass of coal as macromolecules, g/mol;

The relationship between excess adsorption amount and absolute adsorption amount conversion is shown in Eq. (2):

$$N_{ex} = N_{ab} - N_A PV_v / RT \tag{2}$$

In formula: N_{ex} is the amount of excess adsorption of gas molecules in the coal macromolecular structure, moleculars/u.c; N_{ab} is the amount of absolute adsorption of gas molecules in the coal macromolecular structure, moleculars/u.c; N_A is Avogadro's constant, which is taken to be 6.02×10^{23} ; and V_v is the pore volume of accessible pores in the coal macromolecular structure, ml.

Converting the excess adsorption of gas molecules adsorbed by the coal macromolecular structure model to the adsorbed amount ml/g at the standard condition can be expressed as:

$$Q_{ex} = 22400 \times \frac{N_{ex}}{M} \tag{3}$$

In formula: Q_{ex} is the excess adsorption of the modelled coal macromolecular structure, ml/g.

In multiphase adsorption, the adsorption selectivity coefficient $S_{i/j}$ is

an important index for evaluating the competitive adsorption between gases, and the adsorption selectivity of component i to component j in a multicomponent adsorption system is defined as follows:

$$S_{i/j} = \frac{x_i/x_j}{y_i/y_j} \tag{4}$$

In formula: x_i is the molar fraction of component i in the adsorbed phase and y_j is the molar fraction of component j in the gas phase. If $S_{i/j} > 1$, it means that the competitive adsorption capacity of gas i in the mixed component is stronger than that of gas j . The greater the adsorption selectivity is, the better the separation performance of the gas mixture.

3. Results and discussion

3.1. Binary adsorption

3.1.1. Adsorption amount by binary adsorption

The competitive adsorption characteristics among the three adsorbate molecules were simulated by using the Giant Canonical Monte Carlo and Molecular Dynamics methods, and the temperature of the coal seam was selected to be 293.15 K. The binary mixed (CH₄/CO₂, CO₂/H₂O, CH₄/H₂O) components were simulated with different molar volume fractions (0.9:0.1, 0.7:0.3, 0.5:0.5, 0.3:0.7, 0.1:0.9) on the gas-fat coal macromolecule model of competitive adsorption behavior, and the adsorption isotherms are shown in Fig. 4. The adsorption of binary mixed components on coal molecules also satisfies the Langmuir equation, and the fitting results are shown in Table 4–6.

As can be seen in Fig. 4(a), the adsorption of CH₄ and CO₂ gradually

Table 2
Physical properties of CH₄, CO₂ and H₂O.

Gas-type	Shape	boiling /K	Dynamic Diameter /nm	polarization ratio /10 ⁻²⁴ cm ³	dipole moment /10 ⁻¹⁸ esu cm	Quadrupole moment /10 ⁻²⁶ esu cm ²
CH ₄	tetrahedron	111.65	0.38	2.60	0	0
CO ₂	linear	194.65	0.33	2.65	0	4.30
H ₂ O	folded linear	373.15	0.27	1.48	1.86	9.77

Table 3

Adsorption simulation parameter settings.

Setting	Parameter	Setting	Parameter
Task	Fixed pressure	Production steps	2×10^7
Quality	Customized	Charges	Forcefield assigned
Forcefield	COMPASS	Electrostatic	Ewald
Equilibration steps	1×10^7	van der Waals	Atom based
Ewald accuracy	0.001 kcal/mol	Cutoff distance	12.5 Å
Exchange	0.39	Conformer	0.2
Rotate	0.2	Translate	0.2

increased with the increase of pressure, and the adsorption of CH₄ was close to zero when the molar ratio of CH₄ was 0.1, and the saturated adsorption of CO₂ was kept at about 13.61 ml/g even when the molar ratio of CH₄ was 0.9. With the increase of molar volume fraction of CO₂, the saturated adsorption of CO₂ increased from 13.61 ml/g to 22.85 ml/g, and the saturated adsorption of CH₄ decreased from 5.64 ml/g to 1.01 ml/g. At the same temperature, the adsorption amount of CO₂ was significantly larger than that of CH₄ within the molar ratios set in this paper, indicating that CO₂ has an obvious competitive adsorption advantage compared with CH₄, which is consistent with the conclusions obtained by Zhou [7] and Wu [26]. The isothermal adsorption line of CO₂/H₂O binary component is shown in Figure 4(b), with the increase of molar volume fraction of H₂O molecules, the saturated adsorption amount of H₂O rises from 34.67 ml/g to 68.43 ml/g, and the saturated adsorption amount of CO₂ decreases from 13.76 ml/g to 2.50 ml/g. The adsorption amount of the gases is positively correlated with the volume fraction, and the adsorption amount of the gases is close to zero when CO₂: H₂O = 1:9, the adsorption amount of CO₂ was close to zero, and the adsorption amount of H₂O molecules increased gradually with the increase of molar volume fraction, CO₂ was in a weak adsorption position in the process of competitive adsorption in this binary system. The adsorption isotherm of CH₄/H₂O is shown in Fig. 4(c), with the increase of molar volume fraction of H₂O molecules, the saturated adsorption of H₂O molecules increased from 54.92 ml/g to 73.9 ml/g, and the saturated adsorption of CH₄ decreased from 9.04 ml/g to 0.09 ml/g, and the adsorption amount of H₂O was obviously larger than the adsorption amount of CH₄. In terms of competition between CH₄ and H₂O From the perspective of competition between CH₄ and H₂O adsorption, the adsorption process of H₂O molecules is due to the existence of hydrogen bonding, and the interaction energy of hydrogen bonding is much higher than the van der Waals energy of CH₄, and the adsorption amount of CH₄ is close to zero when the molar ratio of CH₄ is lower than 0.3. Compared with the single-component adsorption amounts, CH₄, CO₂ and H₂O are subject to competitive adsorption, and the saturated adsorption amounts are lower than the single-component saturated adsorption amounts. Mixed adsorbent is not an ideal gas, due to the different sizes and properties of the gases, the regions accessible to the adsorbent are

Table 4CH₄/CO₂ binary component Langmuir fitting results.

CH ₄ : CO ₂	CH ₄			CO ₂		
	a/ (ml/ g)	b/ (1/ MPa)	R ²	a/ (ml/ g)	b/ (1/ MPa)	R ²
0.1:0.9	0.45	0.30	0.9945	23.99	0.96	0.9939
0.3:0.7	1.65	0.49	0.9115	21.58	0.90	0.9962
0.5:0.5	2.38	0.55	0.8901	18.92	1.44	0.9957
0.7:0.3	3.30	0.96	0.9105	15.88	1.89	0.9987
0.9:0.1	5.52	0.78	0.9635	13.93	1.31	0.9900

Table 5CO₂/H₂O binary component Langmuir fitting results.

CO ₂ : H ₂ O	CO ₂			H ₂ O		
	a/ (ml/ g)	b/ (1/ MPa)	R ²	a/ (ml/ g)	b/ (1/ MPa)	R ²
0.1:0.9	2.62	1.78	0.6992	81.89	0.65	0.9943
0.3:0.7	6.82	6.16	0.2978	73.65	0.17	0.9971
0.5:0.5	7.85	0.01	0.9976	66.71	0.43	0.9900
0.7:0.3	11.84	0.22	0.9984	61.58	0.54	0.9768
0.9:0.1	14.48	0.51	0.9882	56.85	0.35	0.9943

Table 6CH₄/H₂O binary component Langmuir fitting results.

CH ₄ : H ₂ O	CH ₄			H ₂ O		
	a/ (ml/ g)	b/ (1/ MPa)	R ²	a/ (ml/ g)	b/ (1/ MPa)	R ²
0.1:0.9	0.09	0.55	0.6245	95.41	0.53	0.9936
0.3:0.7	0.27	0.77	0.6684	84.54	0.69	0.9900
0.5:0.5	3.52	0.84	0.8824	80.28	0.67	0.9885
0.7:0.3	6.88	0.87	0.9924	71.10	0.61	0.9865
0.9:0.1	8.64	0.88	0.9961	62.22	0.47	0.9809

different, and for each component, the larger the molar volume fraction, the larger the adsorption amount of the component. The binary adsorption isotherms for different ratios are close to the conclusions obtained by [27–29]. Tables 7–9.

3.1.2. Adsorption selectivity of binary adsorption

In order to study the competitive adsorption behavior in binary mixed components, the selectivity of CO₂/CH₄, H₂O/CO₂, and H₂O/CH₄ binary mixed components during adsorption in coal was calculated for the temperature of 293.15 K and the pressure ranging from 0.01 MPa to 15 MPa, as shown in Fig. 5. In the binary adsorption system, the trend of adsorption selectivity coefficient with pressure is similar, although the value of selectivity coefficient *S* is different, the adsorption selectivity

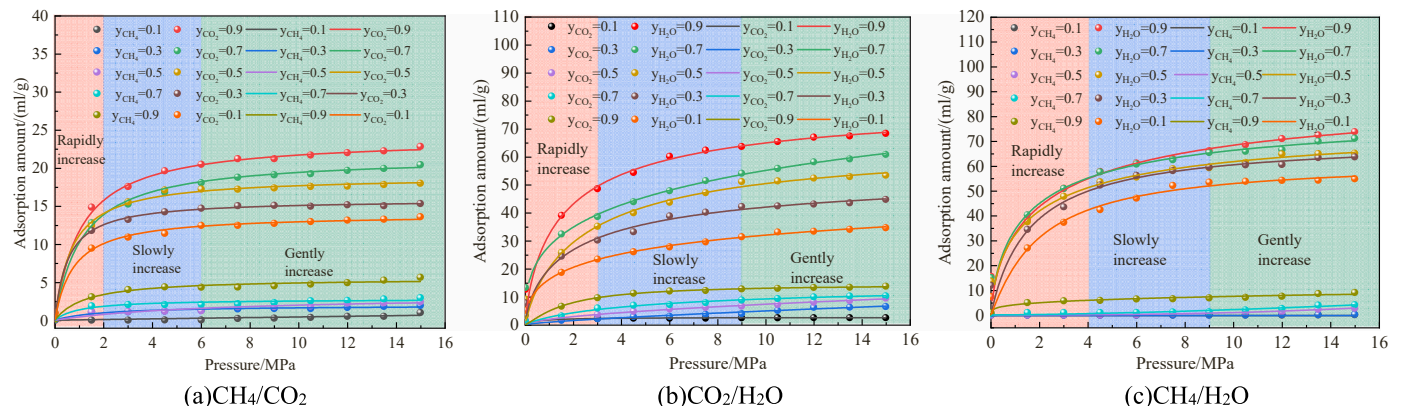
**Fig. 4.** Adsorption isotherms for binary mixed systems with different ratios.

Table 7Total interaction energy of CH₄/CO₂ at different molar fractions.

Pressure/MPa	Total interaction energy of CH ₄ /CO ₂ /kJ·mol ⁻¹				
	0.9:0.1	0.7:0.3	0.5:0.5	0.3:0.7	0.1:0.9
0.01	-303.61	-484.34	-549.60	-634.25	-645.83
1	-1079.58	-1103.00	-1184.55	-1277.8	-1351.08
8	-1217.95	-1237.84	-1380.65	-1455.11	-1539.32
15	-1389.80	-1454.75	-1474.52	-1527.54	-1617.37

coefficient of strong adsorbate gas to weak adsorbate gas decreases with the increase of the equilibrium pressure, and the smaller the molar volume of strong adsorbate gas is, the larger the value of adsorption selectivity coefficient is. For CH₄/CO₂ and CO₂/H₂O systems, the trend of adsorption selectivity coefficients with pressure is relatively similar, and the magnitude of the change is small with the increase of pressure. While the magnitude of the change of adsorption selectivity coefficients with pressure is larger in the CH₄/H₂O system, which indicates that the competitiveness of H₂O for CH₄ is pressure sensitive.

For the CH₄/CO₂ system, when the molar volume fraction of CO₂ increases from 0.1 to 0.9, the adsorption selectivity coefficient at 0.01 MPa decreases from 28.55 to 1.62, and the selectivity coefficient of CO₂ to CH₄ always exceeds 1, which indicates that the adsorption of CO₂ is more competitive than that of CH₄, and that's the reason why CO₂ can drive coalbed methane, and the smaller the molar volume fraction of CO₂ is, the smaller the equilibrium pressure is, and the larger the adsorption selectivity coefficient is, the more favorable the competition of CO₂ to CH₄ is. The smaller the molar volume fraction of CO₂ is, the smaller the equilibrium pressure is, the larger the adsorption selectivity coefficient is, and the more conducive to the competitive adsorption of CO₂ on CH₄. However, high molar volume fraction and high gas injection pressure are favorable to increase the adsorption amount of CO₂, and the appropriate gas injection pressure should be considered during the construction of CO₂-ECBM [30].

Preferential adsorption sites preferentially capture strongly competing adsorbates at low equilibrium pressures and low molar volumes, and the preferential adsorption sites have a large adsorption potential well, but the number of preferential adsorption sites is limited, with the increase of equilibrium pressure and molar volume fraction, the preferential adsorption sites are occupied by the strongly selective gas, and turn to the sub-priority adsorption sites, and the interaction energy decreases, so its adsorption selectivity decreases. Therefore, its adsorption selectivity gradually stabilized with the increase of equilibrium pressure. At low pressure and low molar volume of CO₂, the interaction between CO₂ and the preferred adsorption sites was larger, and these adsorption sites increased the competitive adsorption capacity of CO₂.

The same law applies to the CO₂/H₂O and CH₄/H₂O systems, and in the binary mixed system, the strongly adsorbed gases have greater adsorption selectivity when they have a smaller molar volume. For the CO₂/H₂O system, when the molar volume fraction of H₂O was increased from 0.1 to 0.9, the adsorption selectivity coefficient at 0.01 MPa decreased from 25.60 to 6.64, indicating that the adsorption of H₂O was more competitive than that of CO₂, and the larger the volume fraction of H₂O and the higher the pressure were, the smaller the adsorption selectivity coefficient of H₂O/CO₂ was, indicating that the high pressure and the high molar volume were not conducive to the competitive

nature of H₂O over CO₂, but high molar volume is favorable for adsorption. For the H₂O/CH₄ system, when the molar volume fraction of H₂O was increased from 0.1 to 0.9, the adsorption selectivity coefficient at 0.01 MPa decreased from 105.60 to 1.15, indicating that the adsorption competitiveness of H₂O was stronger than that of CH₄, and the larger the volume fraction of H₂O and the higher the pressure, the lower the adsorption selectivity coefficient of H₂O/CH₄, and the high-pressure and the high molar volume inhibit the competitiveness. In summary, the order of competitive adsorption capacity of the three gases in coal is: H₂O > CO₂ > CH₄.

3.1.3. Probability density distribution of binary adsorption

CH₄, CO₂ and H₂O molecules have the same adsorption sites in Jixi gas-fat coal, and the binary adsorption process occurs the phenomenon of competition for the adsorption sites [31], and the probability density distributions of binary adsorption are shown in Figs. 6–8, which can intuitively reveal the occupation of the adsorption sites by the mixed components.

The probability density distributions of CH₄/CO₂ binary components when adsorbed at different molar volumes at a temperature of 293.15 K and a pressure of 15 MPa are shown in Fig. 6, where the adsorption sites of CH₄ are in blue color and the adsorption sites of CO₂ are in red color, and it can be observed that, with the increase in the molar volume fraction of CO₂, the adsorption sites of CH₄ are gradually occupied by CO₂, and this leads to a gradual increase in the probability density of CO₂. Observing the A region shown in Fig. 6, as the volume fraction of CO₂ increases, the density of CH₄ gradually decreases, the density of CO₂ gradually increases, and when the molar volume of CO₂ reaches 90%, the A region has been completely occupied by CO₂, which becomes the adsorption site of CO₂, and the CH₄ molecules are gradually replaced by the CO₂ molecules in this region. Observing region B, subjected to CO₂ competition adsorption, the adsorption sites in this region are all CH₄ adsorption sites initially, when the CO₂ volume fraction is 30%, they are almost occupied by CO₂, when the CO₂ volume fraction is 50%, four adsorption sites are re-occupied by CH₄, and finally most of the adsorption sites in this region are occupied by CO₂, and the adsorption sites in this region belong to the region where CO₂ and CH₄ repeatedly compete for each other. The adsorption sites in this region belong to the region of repeated competition between CO₂ and CH₄. Except for the circled area for easy observation, the rest of the CH₄ adsorption sites are affected by the competition adsorption of CO₂.

As can be seen from Fig. 7, with the increase of the molar volume fraction of H₂O molecules, the adsorption sites of CO₂ in the regions A and B gradually decrease and the density gradually decreases, and the adsorption sites of H₂O molecules gradually increase and the density constantly increases, when CO₂:H₂O = 0.1:0.9, the CO₂ adsorption sites are almost occupied by H₂O, and there are very few CO₂ adsorption sites. There is a competitive adsorption relationship between CO₂ and H₂O molecules. Compared with the single-component final state probability density distribution, H₂O molecules will compete with CO₂ for adsorption sites, while the adsorption sites of H₂O molecules are less affected by CO₂, and with the increase of the molar volume fraction of H₂O molecules, the adsorption sites of CO₂ are originally occupied by H₂O. The competition between CO₂ and H₂O molecules for adsorption sites is a dynamic and orderly process that is constantly changing.

Observing Figure 8, with the increase of H₂O volume fraction, the

Table 8Total interaction energy of CO₂/H₂O at different molar fractions.

Pressure/MPa	Total interaction energy of CO ₂ /H ₂ O /kJ·mol ⁻¹				
	0.9:0.1	0.7:0.3	0.5:0.5	0.3:0.7	0.1:0.9
0.01	-1115.02	-1383.57	-1507.93	-1804.78	-1888.26
1	-2682.22	-2792.23	-3132.72	-3700.3	-3835.47
8	-3204.65	-3346.58	-3582.51	-4148.44	-4334.03
15	-3678.33	-3995.46	-4179.85	-4448.86	-4566.67

Table 9Total interaction energy of CH₄/H₂O at different molar fractions.

Pressure/MPa	Total interaction energy of CH ₄ /H ₂ O /kJ·mol ⁻¹				
	0.9:0.1	0.7:0.3	0.5:0.5	0.3:0.7	0.1:0.9
0.01	-944.60	-1356.38	-1668.23	-1999.81	-2045.15
1	-2404.54	-3134.89	-3438.08	-3627.43	-3877.90
8	-3199.20	-3524.23	-3614.34	-4117.08	-4439.99
15	-3322.68	-3650.33	-3885.45	-4765.18	-4982.20

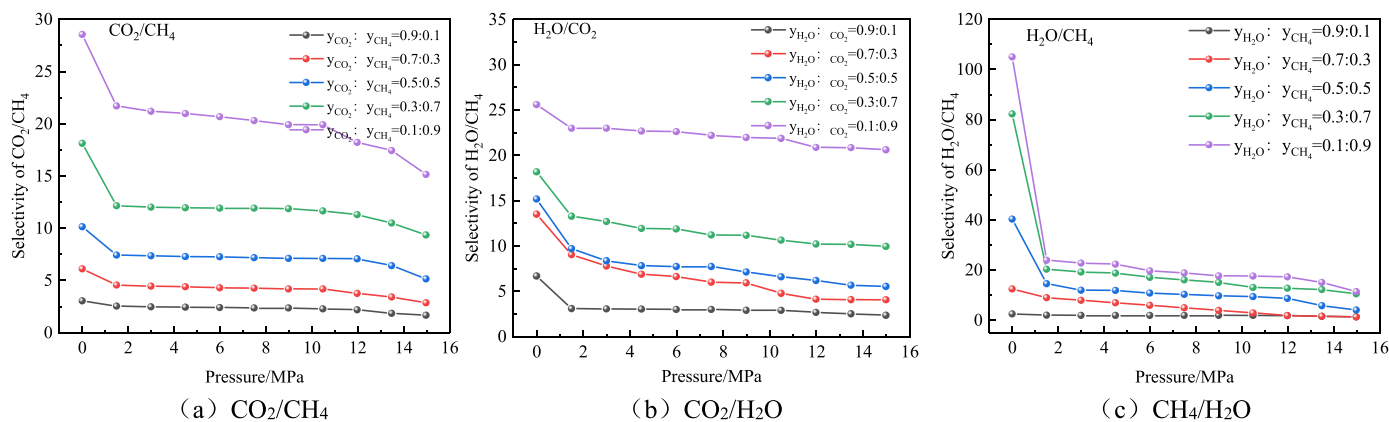


Fig. 5. Variation of adsorption selectivity coefficient with pressure and molar fraction for binary gas mixture.

CH₄ molecules in the B region are gradually occupied by H₂O molecules, and only a small part of CH₄ is left, and the probability of CH₄ is very small at this time, and the adsorption site of CH₄ gradually disappears under the influence of the competitive adsorption of H₂O, and when the volume fraction of H₂O molecules is 90%, all the adsorption sites are almost fully occupied by H₂O molecules, and only a few sporadic CH₄ adsorption sites, and the density is small, while the density of adsorption sites of H₂O molecules gradually increases, and the position distribution is almost unchanged, which is consistent with the adsorption amount law. The competition of adsorption sites exists in the adsorption process of binary mixed gases, with the increase of molar volume fraction of strong adsorbent gas, a part of the adsorption sites occupied by weakly adsorbent gas is gradually occupied by strong adsorbent gas until it is completely occupied; the other part of the adsorption sites is occupied by binary gases alternately due to the competition of adsorption in the adsorption process. Competition for adsorption is a dynamic process.

3.1.4. Potential energy distribution of binary adsorption

In order to study the competition between binary gases for

adsorption sites and analyze the effect of different molar volume fractions on adsorption sites, the potential energy distribution of adsorption under different molar volume fractions of CH₄/CO₂, CO₂/H₂O, and CH₄/H₂O at 15 MPa and 293.15 K was studied, as shown in Fig. 9. Define potential energy peaks as most significant interaction potentials. For the CH₄/CO₂ mixed system, as the volume fraction of CO₂ increases, the most significant interaction energy of CO₂ increases from -39.08 kJ/mol to -34.07 kJ/mol, and the most significant interaction energy of CH₄ decreases from -21.10 kJ/mol to -23.20 kJ/mol; For the CO₂/H₂O mixed system, as the volume fraction of H₂O increases, the most significant interaction energy of H₂O increases from -58.36 kJ/mol to -52.36 kJ/mol, and the most significant interaction energy of CO₂ decreases from -33.23 kJ/mol to -36.16 kJ/mol; For the CH₄/H₂O mixed system, as the volume fraction of H₂O increases, the most significant interaction energy of H₂O shifts from -60.40 kJ/mol to -55.80 kJ/mol, and the most significant interaction energy of CH₄ shifts from -24.45 kJ/mol to -25.71 kJ/mol.

For the CH₄/CO₂ binary mixed system, the potential energy distribution curve of CO₂ shifted to the right and that of CH₄ shifted to the left

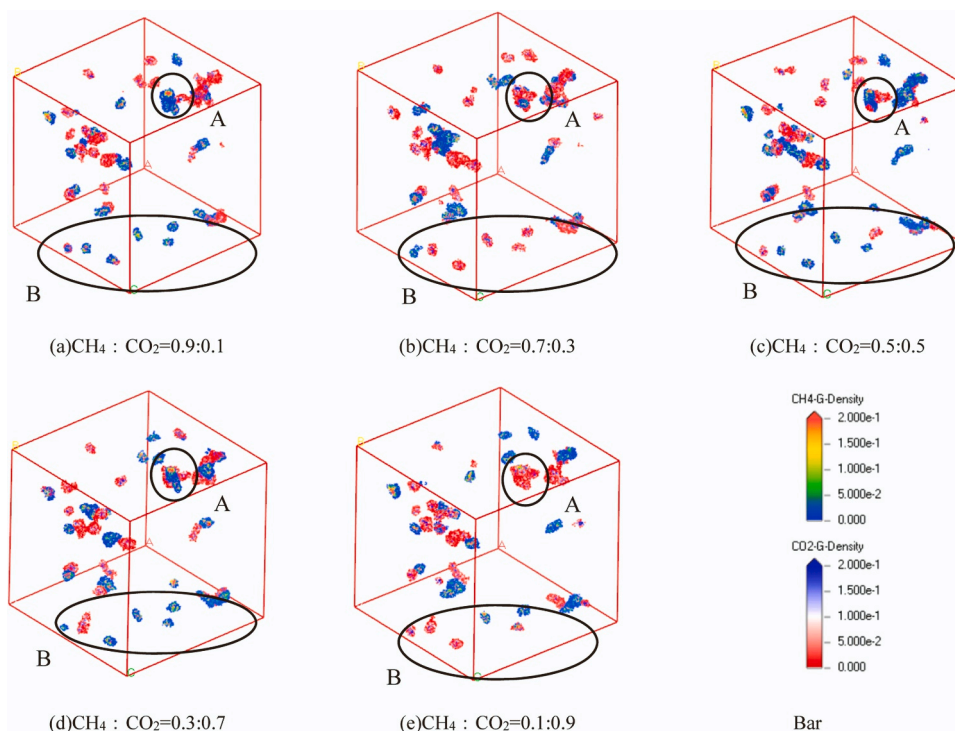


Fig. 6. Probability density distribution of CH₄/CO₂ adsorption in different molar volumes.

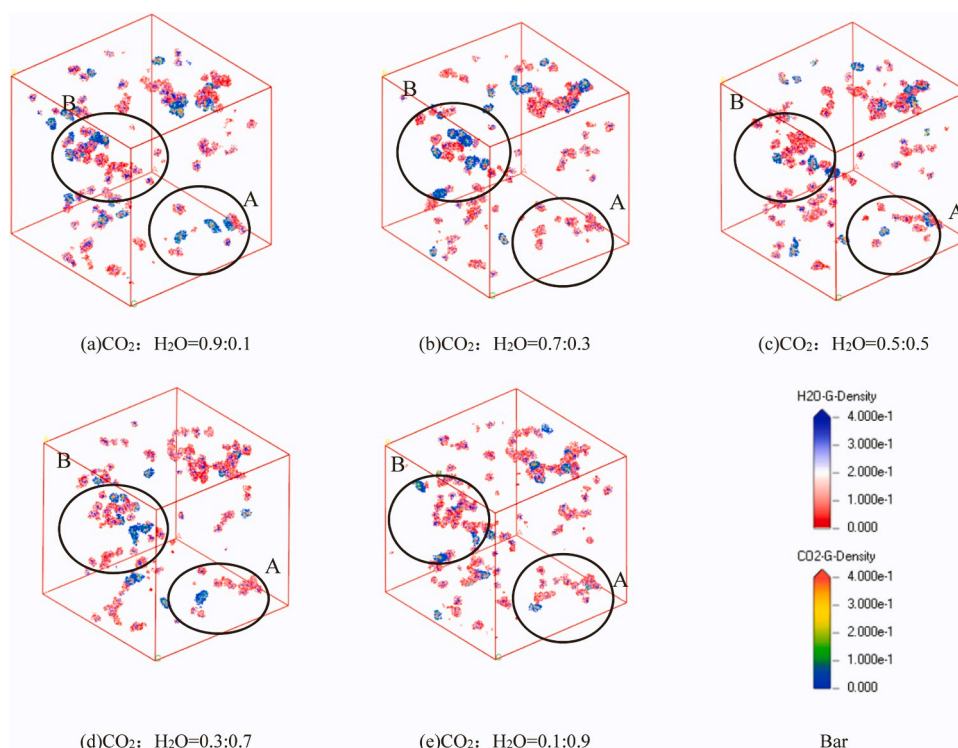


Fig. 7. Probability density distribution of $\text{CO}_2/\text{H}_2\text{O}$ adsorption in different molar volumes.

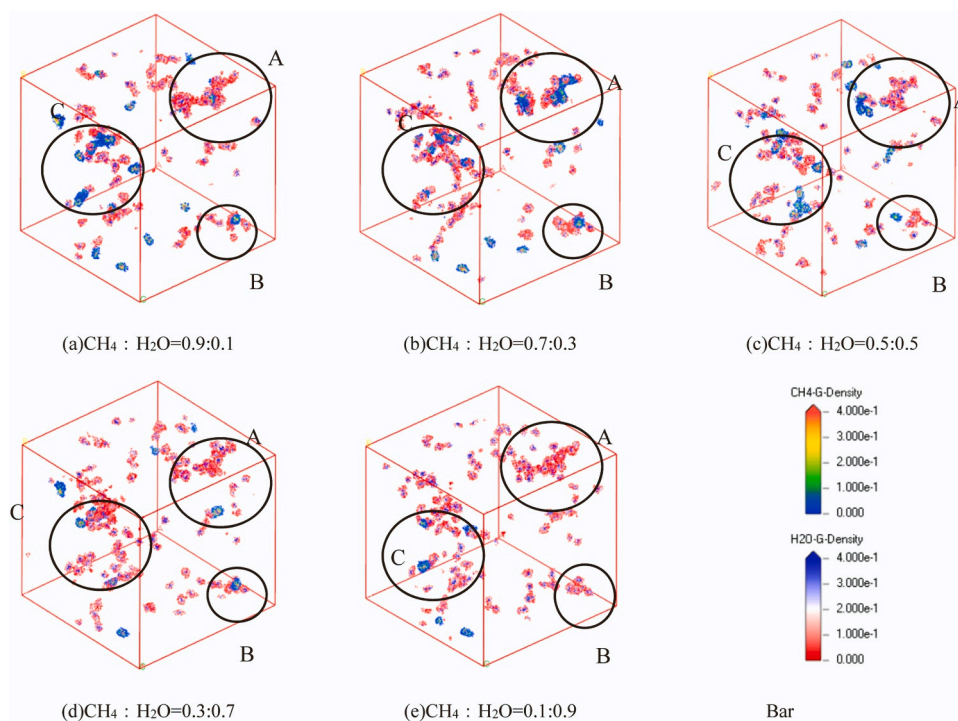


Fig. 8. Probability density distribution of $\text{CH}_4/\text{H}_2\text{O}$ adsorption in different molar volumes.

as the molar volume fraction of CO_2 increased, and the peak of the potential energy distribution curve of CH_4 decreased. With the increase of CO_2 molar volume fraction, the CO_2 molecules were transferred from the preferential adsorption site to the second preferential adsorption site, and the adsorption potential well of the second preferential adsorption site was lower, which led to the rightward shift of the potential energy distribution curve, and the larger the CO_2 molar volume fraction was,

the more the degree of the rightward shift was. The CH_4 potential energy distribution curve gradually shifted to the left as the molar fraction decreased, and the smaller the CH_4 molar volume fraction, the stronger its competitive adsorption capacity and the greater the possibility of occupying the preferential adsorption site, the more it shifted to the left. The adsorption of CH_4 molecules decreases with the decrease of molar volume fraction, and the concentration of each adsorption site in the

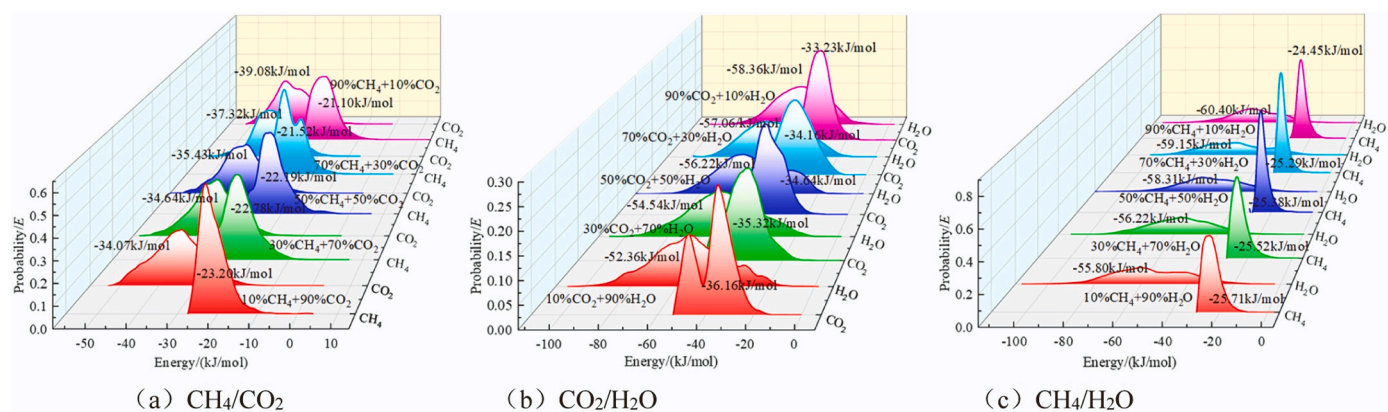


Fig. 9. Potential energy distribution of binary gas mixtures with different volume fractions.

gas-fertilized coal molecules decreases. The overlap of their potential energy distribution curves is the region of competition for adsorption sites, and the more the overlap of their potential energy distribution curves, the more adsorption sites are available for competition between CH_4 and CO_2 , and the more intense the competition is [32]. Similarly, the theory also applies to $\text{CH}_4/\text{H}_2\text{O}$ and $\text{CO}_2/\text{H}_2\text{O}$ systems where the potential energy distribution is affected by competitive adsorption. In the $\text{CO}_2/\text{H}_2\text{O}$ system when the volume fraction of H_2O molecules is 90%, the potential energy distribution of CO_2 shows a bimodal distribution, which is mainly due to the higher volume fraction of molecules, the H_2O adsorption amount is larger, with the increase of H_2O adsorption, H_2O molecules accumulate locally, and water clusters are gradually formed, and there is an interaction between water clusters and CO_2 , which triggers the adsorption of CO_2 by the H_2O clusters. H_2O clusters gradually formed new adsorption sites to adsorb CO_2 . Comparing the potential energy distribution diagrams of CH_4 in binary CH_4/CO_2 and $\text{CH}_4/\text{H}_2\text{O}$ with that of CH_4 monomer gas adsorption, it is found that when the molar volume fraction of CH_4 is smaller, the width of its potential energy distribution curve becomes narrower and the peak value rises compared with that of the monomer potential energy distribution curve, and this phenomenon is especially obvious in the $\text{CH}_4/\text{H}_2\text{O}$ system. This is mainly due to the competitive adsorption of CO_2 and H_2O strong adsorption gases, CO_2 and H_2O occupy the adsorption sites of CH_4 , so that the adsorption space of CH_4 becomes smaller, and the adsorption sites are more centralized, so that the probability of its potential energy distribution rises.

3.1.5. Isosteric heat of adsorption for binary adsorption

The isosteric heat of adsorption is the amount of thermal energy

released during adsorption and reflects the adsorption capacity [33]. Fig. 10 depicts the isosteric heat of adsorption ($-Q_{st}$) for the binary mixed systems of CH_4/CO_2 , $\text{CO}_2/\text{H}_2\text{O}$, and $\text{CH}_4/\text{H}_2\text{O}$ when adsorbed at different volume fractions, and the enthalpy changes ΔH ($-Q_{st}$) of the three binary components are negative, which confirms that the adsorption of the three adsorbate species on gas-fat coal is an exothermic reaction.

The trend of the isosteric adsorption heat of CH_4/CO_2 binary components with pressure is shown in Fig. 10 (a). It can be observed that the isosteric adsorption heat of CO_2 decreases during the low-pressure stage, and then slightly increases with the increase of pressure until the isosteric adsorption heat stabilizes. As the volume fraction of CO_2 increases, the isosteric adsorption heat increases. The isosteric adsorption heat of CH_4 slightly increases with the increase of pressure, and fluctuates slightly due to the influence of CO_2 competitive adsorption. Moreover, when the volume fraction of CH_4 is 0.1, the fluctuation amplitude is the largest. The higher the CH_4 volume fraction, the lower the volatility of its isosteric heat of adsorption. The trend of isosteric heat of adsorption of $\text{CO}_2/\text{H}_2\text{O}$ binary components with pressure is shown in Fig. 10(b). It is observed that the isosteric heat of adsorption of H_2O decreases gradually with increasing pressure from 0.01 MPa to 1.5 MPa, and the lower the volume fraction of H_2O , the greater the decrease. When the pressure is greater than 1.5 MPa, the isosteric heat of adsorption rises gradually with increasing pressure; The isosteric heat of adsorption of CO_2 decreases in the low-pressure phase and increases in the high-pressure phase. The trend of the isosteric heat of adsorption of the $\text{CH}_4/\text{H}_2\text{O}$ binary component with pressure is shown in Fig. 10(c), and it is observed that the isosteric heat of adsorption of H_2O in the low-pressure stage decreases gradually with the increase of the pressure, and the lower the

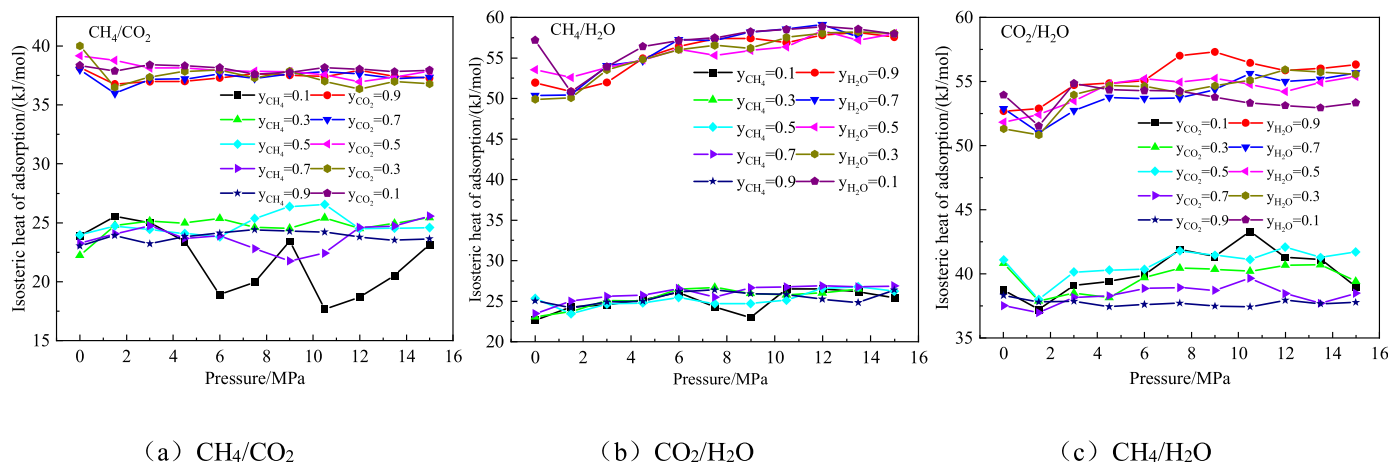


Fig. 10. Isosteric heat of adsorption for binary mixed systems with different volume fractions of adsorption.

volume fraction of H_2O is, the greater the decrease is, and the isosteric heat of adsorption of CH_4 increases slightly with the increase of the pressure. When the pressure is 0.01 MPa, the strongly adsorbate gas shows higher isosteric heat, and the lower the molar volume fraction, the higher the isosteric heat, indicating the stronger adsorption interactions with the coal, and the lower the molar volume fraction of the strongly adsorbate gas, the stronger the competing adsorption selective ability is shown.

Isosteric heat of adsorption of CO_2 and H_2O in the low-pressure stage appears to reduce phenomenon exists, mainly because: in the low-pressure adsorption stage, CO_2 and H_2O preferentially adsorbed in the high-potential energy priority adsorption site, the adsorption site has the largest potential energy, the strongest interaction, and the release of the heat of adsorption is larger, and with the adsorption, the adsorbent material is shifted to the second-priority adsorption site from the priority adsorption site, resulting in the reduction of the heat of adsorption. The smaller the volume fraction, the greater the adsorption heat released during the initial stage of adsorption, and the more significant the reduction in adsorption heat. The smaller the volume fraction of the dominant adsorbed gas, the larger the selectivity coefficient, the more pronounced the competitive effect, the easier it is to enter the high-potential energy adsorption sites, and the higher the heat of adsorption released. When the pressure is greater than 1.5 MPa, the equivalent heat of adsorption of H_2O rises gradually as the pressure increases, and the H_2O - H_2O interactions increase gradually as adsorption proceeds, and the hydrogen bonding interactions between H_2O and H_2O are numerically much larger than the H_2O -coal interactions, which results in a rise in the heat of adsorption [26]. The heat of adsorption of CH_4 shows local fluctuation due to competitive adsorption, and the smaller the volume fraction, the greater the volatility. When the volume fraction of CH_4 is small, the gas selective adsorption selectivity is larger, the gas competition effect is strong, in the process of competitive adsorption, the weak adsorption gas is replaced by the strong adsorption gas, the weak adsorption gas appears desorption phenomenon, the desorption process is heat absorption, resulting in the decline of CH_4 adsorption heat, by the effect of competitive adsorption, the CH_4 molecules are frequently adsorbed and desorbed, resulting in the large fluctuation of the heat of adsorption of the equal amount of adsorption.

3.1.6. Interaction energy for binary adsorption

The interaction energy reflects the stability of the adsorption system, and the larger the absolute value of the interaction energy, the more stable the system is [34,35]. The adsorption process is mainly affected by the interaction energy, and in order to essentially understand the contribution of the interaction energy to the adsorption amount, the interaction energy during adsorption of a binary system at 293.15 K is investigated in this paper, as shown in Fig. 11.

From Fig. 11(a), the interaction energy of the CH_4/CO_2 system

decreases from -303.61 kJ/mol to -1617.37 kJ/mol at 293.15 K when the pressure is increased from 0.01 MPa to 15 MPa. The larger the absolute value of the interaction energy, the easier it is to adsorb [36], and when the pressure increases, the electrostatic and van der Waals energies increase rapidly and then slowly, which is consistent with the increasing trend of adsorption, and indicates that the adsorption sites with larger interaction energies preferentially adsorb gas molecules. In the competitive adsorption process of CH_4/CO_2 binary system, the interaction energy is dominated by van der Waals interaction, which accounts for about 70% of the interaction energy, and the electrostatic energy accounts for about 30% of the interaction energy, and the electrostatic interactions are closely related to the adsorption of CO_2 . The interaction energy of the binary mixed system of CH_4/CO_2 increases with the increase of the volume fraction of CO_2 , which implies that the interaction energy between CO_2 and coal dominates, and the CO_2 has more competitive adsorption ability than CH_4 , so CO_2 volume fraction increases and the total interaction energy increases. As can be seen from Fig. 11(b), the interaction energy of $\text{CO}_2/\text{H}_2\text{O}$ system decreases from -1115.02 kJ/mol to -4566.67 kJ/mol when the pressure is increased from 0.01 MPa to 15 MPa at 293.15 K. The interaction energy of the binary mixed system of $\text{CO}_2/\text{H}_2\text{O}$ adsorbed with different volume fractions is dominated by hydrogen bonding energy and electrostatic energy, which account for about 85% of the interaction energy, and the higher the pressure, the higher the degree of hydrogen bonding energy and electrostatic energy. The total interaction energy of the $\text{CO}_2/\text{H}_2\text{O}$ binary mixture system increased with the increase of the volume fraction of H_2O , which indicated that the hydrogen bonding between H_2O - H_2O dominated the interaction energy of the mixture system.

From Fig. 11(c), the interaction energy of the $\text{CH}_4/\text{H}_2\text{O}$ system decreases from -944.60 kJ/mol to -4982.20 kJ/mol when the pressure is increased from 0.01 MPa to 15 MPa at 293.15 K. The hydrogen bonding interaction accounts for about 95% of the interaction energy of the $\text{CH}_4/\text{H}_2\text{O}$ binary system in the process of competitive adsorption. And the higher the pressure, the greater the hydrogen bonding energy is. The interaction energy of the binary $\text{CH}_4/\text{H}_2\text{O}$ mixture system increases with the increase of the volume fraction of H_2O , indicating that the hydrogen bonding between H_2O - H_2O dominates the interaction energy. Observing the interaction energies of the $\text{CO}_2/\text{H}_2\text{O}$ and $\text{CH}_4/\text{H}_2\text{O}$ systems, when the volume fraction of H_2O is greater than 70%, the total interaction energy of the binary system has a jump in the high-pressure stage compared to the H_2O molar volume less than 70%, and with the increase of the pressure, the adsorption of H_2O gradually increases, and the water clusters are gradually formed [37,38], and the emergence of H_2O - CO_2 and H_2O - H_2O interactions results in the generation of the jump in the total interaction energy.

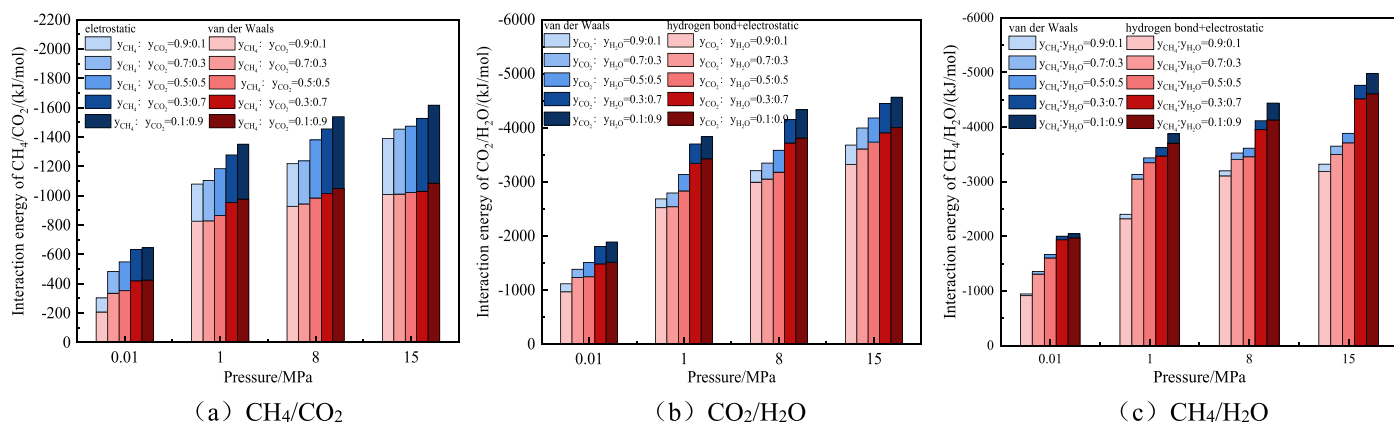


Fig. 11. Interaction energy for competitive adsorption of binary components.

3.2. Ternary competition adsorption

3.2.1. Adsorption by ternary competition adsorption

In order to study the competitive adsorption behavior of three gases, this section studied the adsorption behavior of the ternary mixture ($\text{CH}_4/\text{CO}_2/\text{H}_2\text{O}$) at different molar volume fractions (0.2:0.7:0.1, 0.4:0.5:0.1, 0.6:0.3:0.1, 0.8:0.1:0.1, 0.33:0.33:0.33) at 293.15 K and pressure of 0.01 MPa~15 MPa. Keeping the molar ratio of water constant, it was found that the adsorption rate of CO_2 was greater than that of CH_4 and H_2O , which was mainly due to the larger adsorption amount of H_2O , which made it difficult to reach adsorption equilibrium in a short period of time, whereas the competing adsorption capacity of CH_4 was weaker in comparison with that of CO_2 and H_2O , and the preferential combination of H_2O and CO_2 with the adsorption sites resulted in the decrease of the adsorption rate of CH_4 . The adsorption rates of H_2O and CO_2 decrease with the decrease of CO_2 molar fraction, resulting in a decrease in CO_2 molar fraction and an intensification of competition. H_2O molecules compete with CO_2 molecules at the same adsorption site, which is alternately occupied by the two gases, resulting in a decrease in their adsorption rates. The adsorption capacities of CH_4 , CO_2 , and H_2O at different molar fractions are shown in Fig. 12, and the Langmuir fitting results of the $\text{CH}_4/\text{CO}_2/\text{H}_2\text{O}$ ternary components are shown in Table 10.

As the pressure increases, the adsorption amount of the adsorbate increases. The adsorption isotherms of H_2O and CO_2 at different molar volume fractions are consistent with Langmuir I. When the molar volume of CH_4 is less than 0.4, its adsorption amount is not affected by pressure and is almost zero, and its adsorption isotherm no longer follows Langmuir. When the molar volume fraction of CO_2 is 0.7, during the low-pressure adsorption stage, the adsorption amount of CO_2 is greater than that of H_2O molecules. As the equilibrium pressure

increases, the adsorption capacity of H_2O molecules is greater than that of CO_2 . Within the simulation range, the adsorption amount of CO_2 increased from 8.31 ml/g to 12.6 ml/g, and the saturated adsorption amount of H_2O decreased from 37.81 ml/g to 19.72 ml/g with the increase of CO_2 molar volume. The adsorption capacity of H_2O gradually decreased with the increase of CO_2 molar volume. The adsorption amount of CH_4 decreased from 4.45 ml/g to 0.82 ml/g with the increase of CO_2 volume fraction. At a temperature of 293.15 K and a pressure range of 0–15 MPa, the saturated adsorption capacity of H_2O is greater than that of CH_4 and CO_2 , and the adsorption capacity of CO_2 is greater than that of CH_4 . The adsorption capacity relationship between the three adsorbate is: $\text{H}_2\text{O} > \text{CO}_2 > \text{CH}_4$. Water vapor will seriously reduce the effect of CO_2 to replace CH_4 , in the process of injecting flue gas to store CO_2 to prevent fire, the flue gas should be dried in advance, and in the process of selecting the site for CO_2 storage, the coal beds with lower water content should be selected as much as possible for storage.

3.2.2. Potential energy distribution of ternary competitive adsorption

The potential energy distribution of ternary $\text{CH}_4/\text{CO}_2/\text{H}_2\text{O}$ components under different molar volume fractions is shown in Fig. 13. Competitive adsorption sites can be analyzed through the potential energy distribution. The larger the absolute value of adsorption potential energy, the stronger the interaction, and the more preferential the adsorption sites. As shown in Fig. 13(a), the most significant interaction energy of CH_4 gradually shifts to the left with an increase in CO_2 molar volume fraction, decreasing from -22.83 kJ/mol to -26.18 kJ/mol. The most few interaction energy of CH_4 is located at -23.62 kJ/mol when $\text{CH}_4:\text{CO}_2:\text{H}_2\text{O} = 1:1:1$. The potential energy distribution is bimodal when the volume fraction of CH_4 is 20%. At this time, the molar volume fraction of CH_4 is small, and the adsorption amount of CO_2 and H_2O molecules is large, and it is inferred that the molecules of CO_2 and

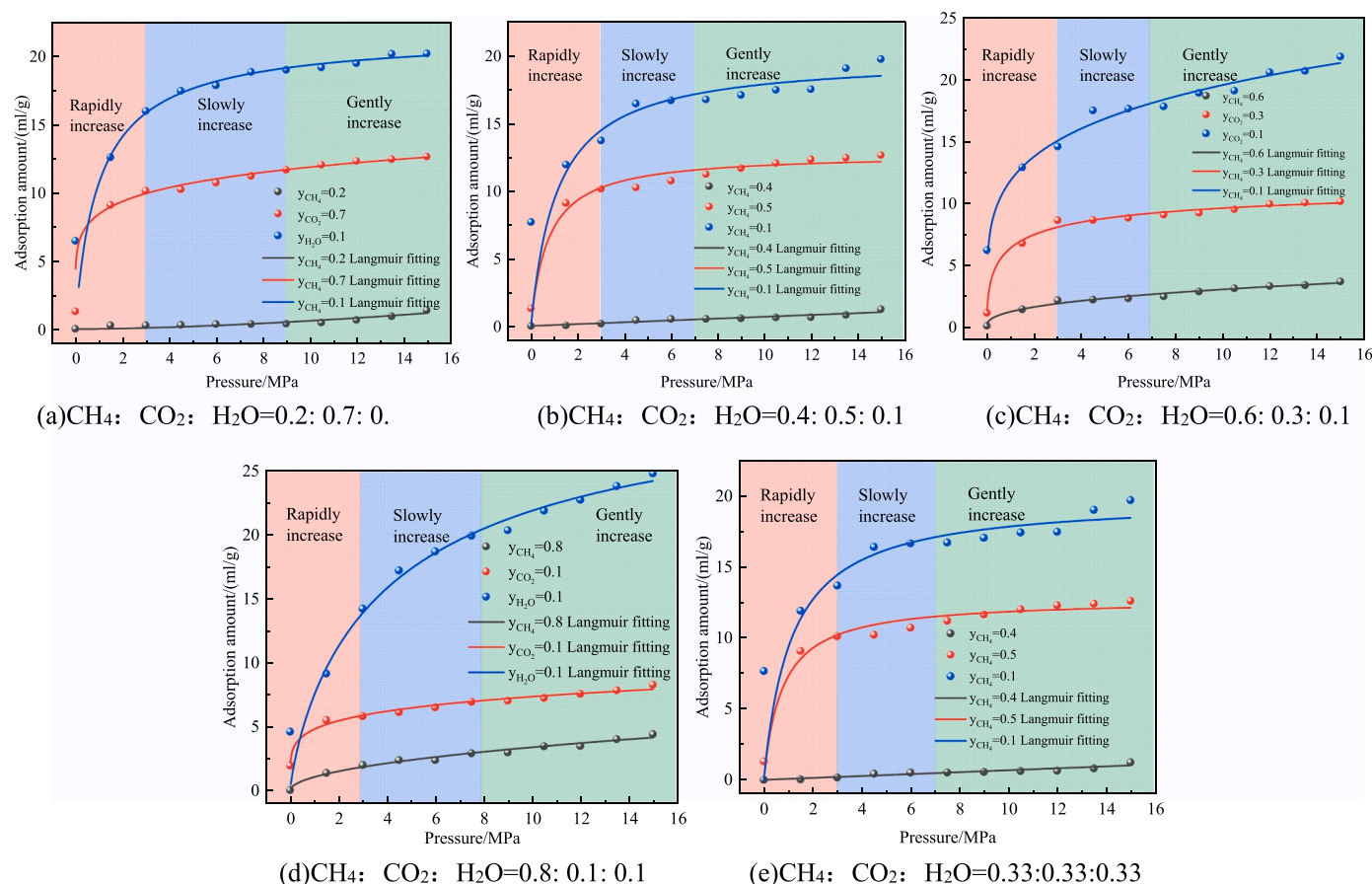
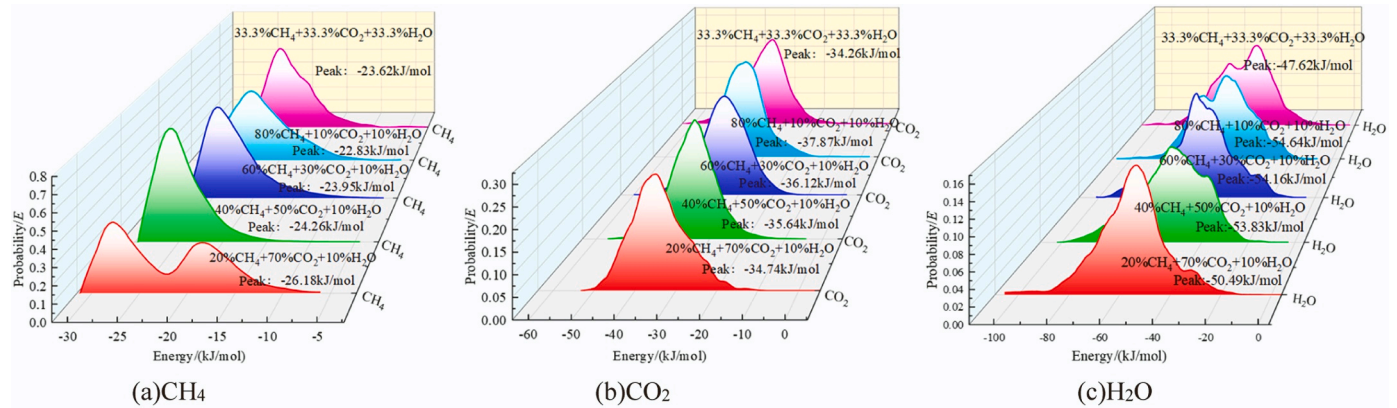


Fig. 12. Isothermal adsorption of with different ratios of ternary components.

Table 10CH₄/CO₂/H₂O ternary component Langmuir fitting results.

CH ₄ :CO ₂ :H ₂ O	CH ₄			CO ₂			H ₂ O		
	a/ (ml/g)	b/ (1/MPa)	R ²	a/ (ml/g)	b/ (1/MPa)	R ²	a/ (ml/g)	b/ (1/MPa)	R ²
0.2:0.7:0.1	0.92	0.48	0.92	13.06	1.59	0.98	22.09	0.61	0.97
0.4:0.5:0.1	1.19	0.62	0.85	12.74	1.34	0.96	23.86	0.89	0.98
0.6:0.3:0.1	3.59	0.76	0.78	10.09	2.89	0.91	25.92	0.63	0.95
0.8:0.1:0.1	4.46	0.89	0.76	8.29	1.32	0.96	33.13	0.29	0.93
0.33:0.33:0.33	1.39	0.93	0.86	8.93	1.33	0.90	48.89	0.39	0.99

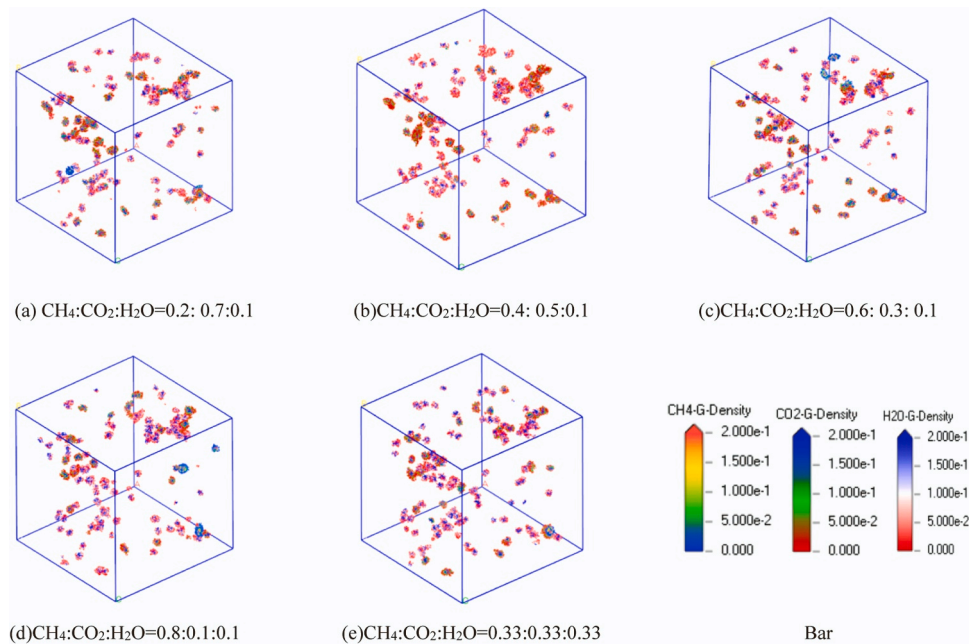
**Fig. 13.** Energy distribution of ternary gas mixtures with different volume fractions.

H₂O form clusters gradually with adsorption, and there are new adsorption sites formed between H₂O-CH₄ and CO₂-CH₄ interact with each other to form new adsorption sites. From Fig. 13(b), it can be seen that the most significant interaction energy of CO₂ gradually shifts to the right, from -37.87 kJ/mol to -34.74 kJ/mol. As the molar fraction of CO₂ increases, this is related to the heterogeneity of coal pore structure; When CH₄:CO₂:H₂O = 1:1:1, the most significant interaction energy of CO₂ is -34.26 kJ/mol. As shown in Fig. 13(c), the potential energy distribution of H₂O molecules gradually shifts to the right as the molar volume fraction of CO₂ increases. The most significant interaction energy increases from -54.64 kJ/mol to -50.49 kJ/mol, and its

adsorption site is mainly affected by the molar volume fraction of CO₂. The CO₂ adsorption amount increases with the increase of CO₂ molar volume fraction. Due to competitive adsorption, the adsorption site of H₂O molecules is gradually occupied by CO₂. When CH₄:CO₂:H₂O = 1:1:1, the most probable interaction energy of H₂O is -47.62 kJ/mol.

3.2.3. Probability density distribution of ternary competitive adsorption

The probability density distribution of competitive adsorption in the CH₄/CO₂/H₂O ternary mixture system at different molar volume fractions is shown in Fig. 14. At any molar volume fraction, H₂O molecules

**Fig. 14.** Adsorption probability density distribution of CH₄/CO₂/H₂O with different mole fractions.

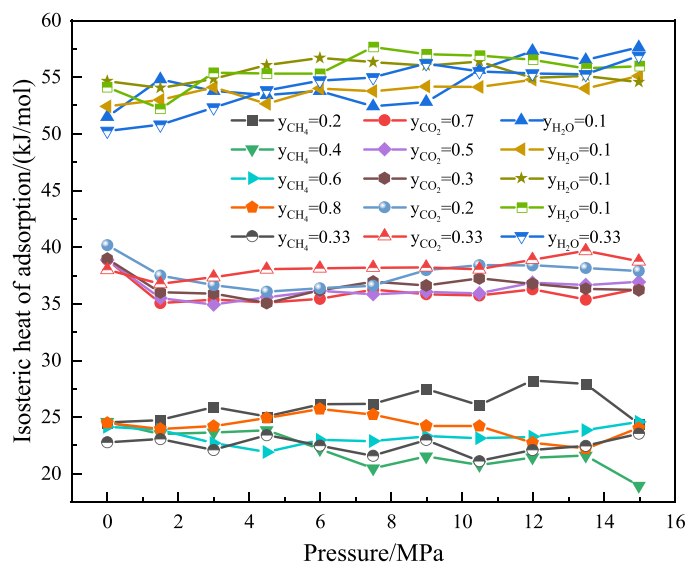


Fig. 15. Isosteric heat of adsorption for CH₄/CO₂/H₂O adsorption at different volume fractions.

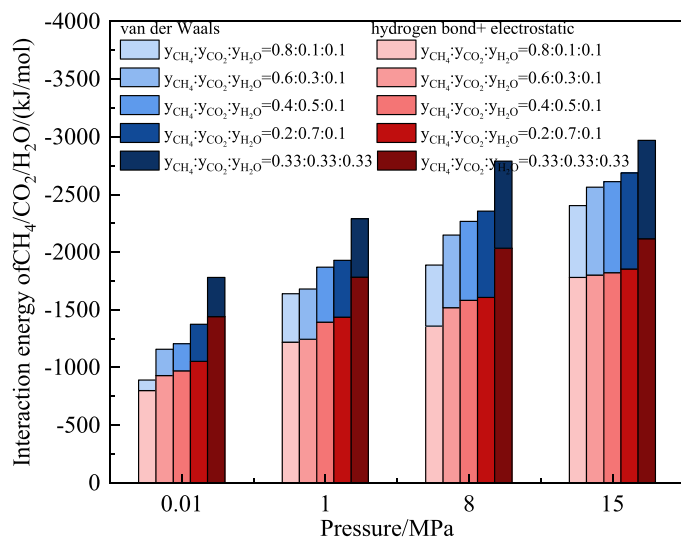


Fig. 16. Variation of ternary component CH₄/CO₂/H₂O interaction energy at different pressure volume fractions.

have the highest adsorption space and density, followed by CO₂ molecules, and CH₄ molecules have the smallest adsorption space and density. As the volume fraction of CO₂ increases, the volume fraction of CH₄ decreases. Due to competitive adsorption, the adsorption sites of CH₄ are occupied by CO₂, resulting in a decrease in the adsorption amount of CH₄. Some adsorption sites of H₂O are also occupied by CO₂, and the adsorption space and density of CO₂ increase, maintaining good consistency with the adsorption amount. Therefore, it can be inferred that the adsorption amount of gas molecules is not only related to their own

properties, but also to the volume fraction.

3.2.4. Isosteric heat of adsorption in ternary competitive adsorption

The variation pattern of the isosteric adsorption heat of CH₄/CO₂/H₂O in the ternary system is shown in Fig. 15. As the volume fraction changes, the isosteric adsorption heat of H₂O fluctuates in the range of 50.24–57.65 kJ/mol, the isosteric adsorption heat of CO₂ fluctuates in the range of 34.88–40.14 kJ/mol, and the isosteric adsorption heat of CH₄ fluctuates in the range of 18.87–28.18 kJ/mol. The isosteric adsorption heat of the three adsorbate molecules, from high to low, is: H₂O > CO₂ > CH₄, consistent with the results of interaction energy and adsorption amount.

In the low-pressure adsorption stage, the isosteric adsorption heat of CO₂ decreases with increasing pressure, then slightly increases with increasing pressure, and gradually flattens out. The larger the volume fraction of CO₂, the smaller the isosteric adsorption heat of CO₂; Due to the volume fraction of fixed H₂O molecules, the adsorption capacity and interaction energy is stronger of CO₂ is greater than CH₄. Therefore, the isosteric adsorption heat of H₂O molecules is more affected by the molar volume fraction of CO₂. When the volume fraction of CO₂ is 0.1 and 0.3, the isosteric adsorption heat of H₂O molecules first decreases and then gradually increases. When the volume fraction of CO₂ is greater than 0.3, the isosteric adsorption heat of H₂O molecules gradually increases with the increase of equilibrium pressure. When the volume fraction of CO₂ is small, CO₂ occupies less preferential adsorption sites, and H₂O molecules are more affected by coal heterogeneity. When the volume fraction of CO₂ is greater than 0.3, the adsorption of H₂O molecules is more affected by the interaction between H₂O-H₂O than by the heterogeneity of coal. The isosteric adsorption heat of CH₄ fluctuates with the increase of equilibrium pressure, and the lower the molar volume fraction of CH₄, the greater its volatility. The adsorption of CH₄ is an exothermic reaction, while the desorption of CH₄ is an endothermic reaction. When the molar volume fraction of CH₄ molecules is small, the competitive adsorption effect is maximum, and CH₄ molecules will frequently undergo adsorption desorption, resulting in greater fluctuations in the isosteric adsorption heat.

3.2.5. Interaction energy of ternary competitive adsorption

The interaction energies of CH₄/CO₂/H₂O at different molar volume fractions and pressures at 293.15 K are shown in Fig. 16, and the total interaction energies of CH₄/CO₂/H₂O at different molar fractions are shown in Table 11.

As the pressure increases, the interaction energy between coal and ternary components gradually increases, which is consistent with the change in adsorption capacity. In the competitive adsorption process of ternary mixed systems, the interaction energy mainly consists of hydrogen bonding energy and electrostatic energy, accounting for about 75% of the total interaction energy. The total interaction energy of the CH₄/CO₂/H₂O ternary mixture system decreases from –2403.45 kJ/mol to –2685.23 kJ/mol with the increase of CO₂ molar volume fraction. When CH₄:CO₂:H₂O = 1:1:1, the total interaction energy is –2967.78 kJ/mol. When the volume fraction of H₂O is 10%, the interaction energy between CO₂ and coal is greater than that of CH₄. As the molar volume fraction of CO₂ increases, the absolute value of the interaction energy continuously increases with the increase of CO₂ volume fraction. Moreover, when CH₄:CO₂:H₂O = 1:1:1, the absolute

Table 11

Total interaction energy of CH₄/CO₂/H₂O at different molar fractions.

Pressure/MPa	Total interaction energy of CH ₄ /CO ₂ /H ₂ O /kJ·mol ⁻¹				
	0.8:0.1:0.1	0.6:0.3:0.1	0.4:0.5:0.1	0.2:0.7:0.1	0.33:0.33:0.33
0.01	-889.96	-1157.83	-1205.44	-1373.35	-1779.02
1	-1639.4	-1679.62	-1868.89	-1927.89	-2288.39
8	-1887.08	-2147.84	-2265.84	-2355.05	-2789.25
15	-2403.45	-2563.01	-2609.46	-2685.23	-2967.78

value of the interaction energy increases significantly compared to the former. The hydrogen bonding interaction between H_2O - H_2O is numerically higher than van der Waals energy and electrostatic energy, The larger the molar volume fraction of H_2O molecules, the greater the absolute value of their interaction energy.

4. Conclusion

(1) The adsorption isotherms of the binary mixture components conformed to the Langmuir model, and the adsorption amounts were $\text{H}_2\text{O} > \text{CO}_2 > \text{CH}_4$, from largest to smallest. CO_2 has a competitive advantage over CH_4 , H_2O has competitive adsorption over CO_2 , and the adsorption amount of each gas component increases with the increase of molar volume fraction. The smaller the molar volume and the lower the equilibrium pressure of a strongly adsorbed gas, the higher the adsorption selectivity coefficient. Strongly adsorbed gases move to adsorption sites with higher interaction energies as the molar volume fraction decreases, and weakly adsorbed gases move to weakly interacting adsorption sites as the molar volume fraction increases. When the pressure is 0.01 MPa, the strongly adsorbate gas shows higher isosteric heat, and the lower the molar volume fraction, the higher the isosteric heat, indicating the stronger adsorption interactions with the coal, and the lower the molar volume fraction of the strongly adsorbate gas, the stronger the competing adsorption selective ability is shown. Competition for adsorption is a dynamic process.

(2) The interaction energy in the CH_4/CO_2 system is mainly van der Waals energy, accounting for about 70%, while the rest is electrostatic energy, mainly derived from the adsorption of CO_2 . The interaction energy in the $\text{CO}_2/\text{H}_2\text{O}$ system is mainly hydrogen bonding energy and electrostatic energy, accounting for about 85%. The interaction energy in the $\text{CH}_4/\text{H}_2\text{O}$ system is mainly hydrogen bonding energy, accounting for about 95%. In binary mixed systems, the interaction energy increases with the increase of the molar fraction of strongly adsorbed gases in the binary system.

(3) For the ternary mixed system of $\text{CH}_4/\text{CO}_2/\text{H}_2\text{O}$, as the molar volume fraction of CO_2 increases, the saturated adsorption amount of CH_4 and H_2O show a decreasing trend, and the most significant interaction energy of CH_4 gradually decreases, while the most significant interaction energy of CO_2 and H_2O gradually increases; In the low-pressure adsorption stage, the isosteric adsorption heat of CO_2 decreases with increasing pressure, then slightly increases with increasing pressure, and then gradually stabilizes. And the larger the molar volume fraction of CO_2 , the smaller its isosteric adsorption heat; The isosteric adsorption heat of CH_4 fluctuates with the increase of equilibrium pressure, and the lower the molar volume fraction of CH_4 , the greater its volatility. In the competitive adsorption process of the ternary mixed system, the interaction energy is mainly composed of hydrogen bonding energy and electrostatic energy, accounting for about 75% of the total interaction energy. The total interaction energy decreases from -2403.45 kJ/mol to -2685.23 kJ/mol with the increase of CO_2 molar volume fraction.

CRediT authorship contribution statement

Jinzhang Jia: Methodology, Supervision, Conceptualization, Funding acquisition. **Yinghuan Xing:** Conceptualization, Investigation, Writing- original draft, Writing-review & editing. **Bin Li:** Data curation, Visualization, Software. **Peng Jia:** Formal analysis. **Yumo Wu;** **Qiang Yang;** **Dongming Wang:** Data curation. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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